Problem 4: Let's consider an infinite sequence of digits constructed of ascending powers of 10 written one after another. Here is the sequence: 1101001000... You are to find out what digit is located a the definite position of the sequence. (Timus. 1209)

Solution:

1	2	3	4	5	6	
1	10	100	1000	10000	10000	....
↓	↓	↓	↓	↓	↓	
1	2	4	7	11	16	

If we observe the position of the digits with '1' in their placeholder, they are actually the values of (triangular number +1). For example, in case of 7, 1 has three zeroes (0) with it and total number of digits is 4. And the summation of from 1 to 3 (1 is excluded, only number of 0s are counted) is $\frac{3(3+1)}{2} = 6$ and $6+1 = 7$ where the digit 1 is.

$\therefore$ 1s are located at (triangular number + 1) th places or, $\left(\frac{n(n+1)}{2} + 1\right)$ th digits are 1s and here n starts from 0.

Prerequisite:

1. Same as problem no. 1.